



University of Greater Manchester

Strategic analysis and application of global Supply Chain Practices: Inditex's hybrid sourcing for resilience

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Submitted by:
Ly Gia Han
ID: lgh4ocd

**Tutor: Husain Munther Jawad
Muhammad Asif Khan**

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INDITEX

1. Introduction

The company is an ultra large fashion retailer in the world with its headquarters in Arteixo, Spain and the parent company of the brands involving: Zara, Pull&Bear, Massimo Dutti and Bershka. The Inditex's business model can be characterized by its highly market- responsive vertically integrated supply chain that allows the company to transform the emergent fashion trends into on-store ready products within as short of 2-3 weeks. During the fiscal year 2024, Inditex has registered the highest revenues in the whole history of the company, standing at 38.6 billion Euros, which is an annual growth of 7.5 percent, and impressive operating margins and cash flow (Inditex, 2025). The performance stands out especially against the current global supply chain destabilization with the effects of the pandemic still being felt and with instability in shipping in top trade routes. Even more remarkable about Inditex is its high degree of verticalization, centralized distribution, and robust utilization of real-time information, including RFID technology. These abilities help the company minimize lead times, stock turnover, and match demand and reduce markdowns all which directly lead to profitability.

But the recent years have changed the risk profile of supply-chain across the globe. It has proven that long and complex offshore supply chains are highly vulnerable due to the COVID-19 pandemic, the Ukraine war, Red Sea disruptions, and the emerging geopolitical tensions. These shocks have led to delays in shipping, ship rerouting, rising freight prices, and some companies have even started to contemplate reshoring or near-shoring to save themselves. The decades-old proximity model developed by Inditex across that case then interestingly applies to how a fashion company can remain durable in times of global shocks.

The modern international business environment is characterized by one of the states, permacrisis, a word developed to refer to a long period of instability and insecurity. In the case of worldwide supply chains, it is not an eventual burst, but a wave upon a wave of shocks: geopolitical turmoil as with the conflict between Russia-Ukraine and the crisis in the Middle East, climate change increasingly disruptive like the Panama Canal drought, regulatory changes that are more sustainable (the Carbon Border Adjustment Mechanism of the EU). SCM has outlived its operational mandate in this volatile environment to emerge as the most important strategic role of multinational

corporations. There is no longer a need to focus on the cost reduction but the key force of competitive advantage, resilience and survival.

This report is a critical analysis of Inditex and how it can continue with its Quick Response (QR) model within the context of the challenges it will face in its business processes in 2024-2025. Unlike other competitors that have a weak supply chain that is enforced by the need of the long supply chain, Inditex endorses the strength of agile sourcing and a tactical mix of offshore and near-shore sourcing.

The article is dedicated to the potential of the vertical integration of Inditex and implementation of real-time data to support its competitive advantage over extremely fast-changing fashions rivals like Shein. It highlights the tension between agility in operations especially in airfreighting through Red Sea blockades and ambitious decarbonization targets. By integrating FY2024/2025 financial statements into the ongoing forward-thinking SCM theories, the report analyses the opportunity given to Inditex to be able to face the fear and still remain at the top of the market despite the issue of survival in the dynamic era of sustainability.

2. Strategic role of SCM in modern organizations

In the turbulent economic environment of the mid-2020s, Supply Chain Management has lost its reputation that has long existed as a historical cost centre or a back-office operation. Nowadays, it finds its core at the center of corporate strategy. In the case of global retailers such as Inditex, SCM is not just a matter of delivering boxes in Factory A to the Store B; it is the central nervous system of the organization, profitability, brand image and existence. This change could be explained by the perception that under a condition of permacrisis, the geopolitical instability in the Red Sea, fluctuation in the cost of raw materials, and the pressing climate imperative, supply chain competes, not only among companies (Christopher, 2016). This is not simply because of the fashion designs that Inditex has dominance; it is because it can implement those designs in a supply chain that is more resilient, flexible and faster than its competitors can implement.

2.1 Supply Chain as a strategic resource (RBV Analysis).

In order to fully realize why Inditex does better than others such as H&M or Gap, we need to employ the RBV. According to this academic framework, sustainable competitive advantage is a result of Valuable, Rare, Inimitable, and Non-substitutable (VRIN). This model suits the supply chain of Inditex, where it has made logistics to be a strategic weapon.

Valuable: The core quality of the supply chain in Inditex is speed, which is also referred to as Rapid-Fire Fulfillment (Aftab et al., 2018). As traditional retailers are trying to achieve 6-month lead times, and their tied up capital to inventory that may not move, Inditex can design something and put it on shelf within less than three weeks. This

pace is economically useful in that it lowers lavishly the requirement of taking down. By producing what the customers want at the moment, then you can sell it at full-price. According to industry statistics, Inditex is selling a much large percentage of merchandise at full retail than the industry average of once every 60, which has a direct positive effect on gross margins to approximately 58% (Inditex, 2024).

Rare: In a world where outsourcing has become obsessed with low bidder, the Inditex strategy is an exception. It has got a strong level of Vertical Integration with major processes of production like fabric dyeing, pattern cutting as well as logistic centers in Spain. Majority of competitors, including the emerging giant that is Shein entirely rely on third party workshop networks. Having the brains of the production process (design and logistics) and only outsourcing the muscle (sewing), Inditex preserves the quality and timeliness that is difficult to regulate in the fashion industry.

Imitable: There are many competitors who have attempted to imitate Zara over the decades but that did not work since the system is socially convoluted. Not only does it involve owning factories, but it is also about the qualitative feedback of thousands of store managers to the designers in Arteixo who then also coordinate with manufacturers locally to adjust the production in the middle of the season. This tacit knowledge which is the human software of the supply chain is something not easily achievable through the purchase of improved software or outsourcing a consulting firm (Barney, 1991).

Non-substitutable: Although the speed of the processes of such ultra-fast fashion players as Shein is remarkable, they do not have the physical infrastructure to provide an actual omnichannel experience. The supply chain of Inditex enables the customers to make purchases online and collect them at a store, or to make purchases online and have it delivered by a local store to their stocks. This is the capability that is driven by their own SINT (Integrated Stock Management System) which generates a level of service that can not be replaced by purely digital or purely physical competitors.

2.2 SCM financial engine the negative working capital cycle.

The effects of the supply chain on the cash flow are one of the most essential, and yet commonly disregarded strategies of Inditex. Many traditional retail firms make payments to their suppliers several months before they can even sell the commodities to the customers and their inventory can be holding huge sums of money. Since its inventory turnover is so great, sometimes more than 12 times annually compared to industry average of 3 to 4-times, it markets the clothes to its customers before having to pay its suppliers (Inditex, 2023). This creates a Negative Operating Working Capital cycle. The supply chain is a good funding tool. Inditex does not need to borrow funds through the banks to open new stores or invest into digital transformation, the cash flow generated through the efficient supply chain is used. This financial self-sufficiency is an immediate response to SCM effectiveness and it confirms the fact that logistics

is no plain truck and no ship game, it is a matter of capital structure and financial fitness.

2.3 Dynamic capabilities: Sensing, Seizing and Transforming.

In addition to fixed resources, Inditex has better Dynamic capabilities (Teece, 2007). The most important strategic asset in a volatile world is the aptitude to change.

Sensing: Inditex is not dependent on crystal balls or long-range forecasting. Its supply chain is capable of listening. At the end of every day, the managers of stores are reporting not only what was sold, but what the customers requested and could not find. Was it in the color green and not blue, they desired that blazer? This is granular data, which is fed back to HQ in real time. The supply chain is like a mega sensor that picks changes in the consumer taste weeks prior to its manifestation in conventional market research.

Seizing: Since Inditex obtains about 50-60% of its product in the local markets (Spain, Portugal, Morocco, Turkey) and not in remote Asian plants, it can take action on such information instantaneously (Inditex, 2022). In case the "sensing" mechanism identifies a trend among floral dresses, the "seizing" mechanism (proximity manufacturing) will be able to manufacture and supply them within 15 days. The competitors who use snail boats provided by the Asian manufacturers simply cannot grab such passing chances.

Transforming: The company has been constantly restructuring its assets to address new threats. The recent decision to invest in new logistics automation (logistics expansion plan 2024-2025) in the amount of €900 million shows the intention of the effort to transform the conditions of physical infrastructure to obtain the growth in the future (Inditex, 2024). During the 2024 crisis in the Red Sea that affected the transportation options of their products, switching between transport modes and reliance of its Turkish suppliers demonstrated that Inditex had a resilience-building capacity to reorganize its operations in real-time, and made it reliable.

3. Application of Supply Chain theories and frameworks

The way to critically analyze the resilience of operations of Inditex is to go beyond any superficial accounts of what is meant by fast fashion and dissect the theoretical framework that makes it possible. Inditex has not got a moving fast thing, it has a structural alignment between product features and supply chain design. It uses four superior theoretical models, namely the Matrix of Fisher, The Decoupling Point (Postponement), the supporting (Transaction cost economics) (TCE) and the Real Options Theory to examine how the company manages to reduce risk and maintain a competitive edge in a volatile global market.

3.1 Matrix of Fisher The Hybrid Sourcing Architecture.

The model of matching the supply chains to products developed by Marshall Fisher in 1997 is the argumentative thesis that a mismatch of the character of product and the supply chain design is the main reason behind the failure of supply chains. The products that Fisher compartmentalizes are of the two types:

- **Functional Products:** The products with stable demand, extended life cycle (more than 2 years) and low profit margin (5-20%). The examples are simple white t-shirts or socks.
- **Innovative Products:** Products that have uncertain demand, short lifecycle (3 months to 1 season), and have high margins (20-60%). Among these are stylish printed dresses or seasonal wearing.

Fisher assumes that high utilization and cost minimization supply chain is necessary in physically efficient products (Equivalent to Physically Efficient), and speed, buffers, and flexibility, supply chain is needed in innovative products (Equivalent to Market Responsive).

The Strategic Application at Inditex: The Bifurcated Model The genius of Inditex was that it did not make the decision between one and the other. Contrary to any of its competitors, including Primark (an entirely efficient/low cost model) or luxury houses (an entirely responsive/high cost model), Inditex uses Hybrid Sourcing Model that strictly defines the segments of its supply chain according to the Fisher criteria.

The Efficient Chain (Asian Sourcing): In case of "Functional" products (i.e. classic menswear, denim, basics), Inditex employs Asian suppliers (China, Bangladesh, India, Pakistan). In this case, lead times are longer (3-6 months) and it requires minimisation of the Total Landed Cost. They are also stable in terms of demand and as such, chances of stockouts and obsolescence will be minimal.

The Responsive Chain (Proximity Sourcing): In the case of an "Innovative" product, Inditex applies its Proximity Cluster (Spain, Portugal, Morocco, Turkey). The manufacturing cost in this location (20-30 premium to that of Asia) is high; however, the Total Cost of Ownership is lower due to the ability of the supply chain to remove deep discount offers (markdowns) as the supply is matched with the demand in real-time.

Table 3.1: Inditex's Application of Fisher's Matrix (Comparative Analysis 2024)

Feature	Efficient Chain (Applied to Basics)	Responsive Chain (Applied to Trends)	Theoretical Justification

Sourcing Region	Asia (China, Bangladesh, Vietnam)	Proximity (Spain, Portugal, Morocco, Turkey)	Cost vs. Speed Trade-off: High labour cost is acceptable for speed in innovative goods.
Product Type	Functional (e.g., Jeans, T-shirts)	Innovative (e.g., Viral trends, Blazers)	Fisher (1997): Match chain type to demand uncertainty.
Lead Time	3–6 Months	2–4 Weeks	Agility: Responsiveness requires geographic proximity to reduce transit buffers.
Inventory Strategy	High Volume / Low Margin	Low Volume / High Margin	Risk Management: Bulk produces only what is predictable.
Transport Mode	Sea Freight (95%+)	Road (Europe) / Air (Intercontinental)	Sustainability vs. Speed: Air freight is reserved for high-margin, time-sensitive goods.
% of Sourcing	Approx. 40-45%	Approx. 50-55%	Hybridity: Balances the portfolio risk.

(Source: Synthesised from Inditex Annual Reports 2023-2024 and Aftab et al., 2018)

3.2 Customer order decoupling Point (CODP) and postponement.

CODP is the point where the value chain is connected to a particular customer order. Below the CODP, the supply chain can be enhanced as Push (forecast-driven) above the CODP and Pull (demand-driven) below the CODP. The Form Postponement (delayed differentiation) is a strategy that is aggressively used by Inditex to manipulate the CODP. Traditional fashion model The CODP will be in the retail store (manufacturing of items will be complete months before). Inditex pushes the CODP all the way to the manufacturing phase.

Mechanism of Action:

- Inventory in "Greige": Inditex has large quantities of its raw material inventory in its Arteixo distribution centres in the form of "greige" (undyed/finished) fabric.
- Delayed Commitment: It would not commit itself to colouring this cloth Red or Blue until real time sales information in the stores has confirmed the trend.
- The Pull Trigger One a trend is being perceived (say, Red floral dresses are selling out in London), then greige material is dyed and cut and sewn within the radius cluster within days.

3.3 Organizational governance and process Flow

The supply chain of Inditex provides an interesting confirmation of the application of Transaction Cost Economics (TCE) by Oliver Williamson, who challenges the common practice in the industry of outsourcing. Competitors such as H&M enter into portfolios of external contracts; whereas Inditex maintains an industrial operation; in which they exercise high Vertical Integration to control the Iron Triangle of Speed, Quality and Cost. This type of structure is necessitated by the desire to reduce uncertainty and safeguard Asset Specificity. In a hyper-volatile market, prohibitive costs in coordination and negotiation would be involved in entering into agreements with an external supplier to have short-lead times orders with the aim of producing high and rapid orders; hence, by acquiring capital-intensive processes like design, cutting, and dyeing, Inditex would be internalizing the uncertainty. In addition, vertical integration safeguard proprietary technology such as the Inditex Open Platform (IOP) and RFID technologies, which means that one specific asset such as these is used effectively without the threat of being held up by technological lags as an external business partner.

The suggested governance allows implementing the so-called "Leagile" (lean + Agile) paradigm introduced by Christopher and Towill. Inditex finds the solution to such a dilemma between efficiency and responsiveness through decoupling of the chain, i.e. it uses the Lean principles as an upstream (to the raw materials and basics) company (sourcing in Asian markets, to gain the economies of scale) and the Agile downstream operation (where finishing and distribution is required). Inditex develops a logistical pulse: shipments twice a week with no consideration of the load factor unlike purely lean managers who intend to wait until they have full truckloads to reduce their costs, time rather than transport efficiency is a value that counts. This differentiation enables the company to be a low cost player on commodities but a high-speed player on fashion trends which could not have happened without the control that vertical integration provided.

3.4 Strategic resilience and theoretical synthesis: real options and competitive advantage.

In addition to operational governance, the Inditex uses the Real Options Theory in redesigning sourcing strategy into a financial defense against geopolitical instability. The company has successfully established a portfolio of options by keeping its supplier base dispersed in Spain, Turkey, China and India so that it has a right to, but not an obligation to change the volume of production. This was observed in the 2024, 2025 crisis of Red Sea; competitors had to contend with stockouts, Inditex had exercised its right to divert its volume out of Asia into near-shore centers in Turkey and Portugal. Though it does cost Europe a "premium" in terms of higher wages, this is considered to be not a cost inefficiently, but rather it is a premium that is paid as an insurance policy against shocks in supply. Inditex is paying capacity slack in close markets to

provide continuity and this justifies the fact that cost of the option will be much less than the risk of losing sales.

4. Critical assessment of response to unpredictable events (2024–2025)

4.1 The disruption: Anatomy of the Red Sea Crisis

The fashion industry experienced an abrupt, seriously damaging and essentially structural shock. The established logistics model designed for mass-market apparel utilizes the Asia to Europe sea corridor to ship raw material like greige fabrics and yarns and finished basics. The successful closure of this route has led global carriers to make use of the Cape of Good Hope around the southern tip of Africa to access Asian and European ports. This implies a fundamental change in the physics of the supply chain in 3 dimensions.

- **Delay in transit time:** This diversion increased standard voyage time from Asian ports to European hubs by about 10 to 14 days. This is a two week delay for a "push" retailer with a six-month lead time that can be absorbed through safety stock. For a "fast fashion" retailer like Zara where the value proposition is built around fleeting trends and speed-to-market, this is a death sentence. Within Inditex's business model a micro-trend might only last for 4 to 6 weeks, and a 14 day delay means the inventory becomes obsolete before it hits the stockroom, and full-price sales become 100% markdowns.
- **Reduced capacity:** Longer voyage times also effectively curtailed global container shipping capacity by about 9% as vessels were tied up longer to do the same rotation, resulting in fewer empty containers available in Asian origin ports, which caused a supply-side shock that further compounded the delay.
- **Cost Inflation:** The shock created a freight rate inflation. Spot rates on the Asia-Europe route were almost five-fold in the first quarter of 2024 as the effects of capacity scarcity and an astronomical war risk insurance surcharge conspired. The inflationary pressure on the gross margins of retailers that were unable to offset it on the consumer.

The situation was complicated by a concurrent disturbance in the Western Hemisphere: a Panama Canal drought, which limited the amount of ships crossing each day. Thus, the combined breakdown of both global choke points created logistical disagreements of so far uncommon dimensions. Inditex recognized that in the economy of attention and newness, there was a currency that was more valuable than freight rates time, while competitors faced with this blend of factors chose to put up with the delays they were facing in order to maintain margin.

4.2 Logistics response: The strategic "air bridge"

When the Red Sea close seasoned and ships had to go the long route around the Cape of Good Hope, the majority of fashion houses attempted to wait it out, reducing

prices, cancelling orders or just accepting that spring collections may be received in summer. Inditex did the opposite. In days the company canceled its freight playbook and paid the freight money, gambling on the additional cost of transport would be less than the lost sales or old stock. It was a classic Real Options thinking: the opportunity to switch between the sea and air was an option, and rather exploit the opportunity when the strike price (greater freight bills) went lower than was anticipated to be lost in the lost margin. The scale of customs figures is:

During the quarter between September 2023 and August 2024, Inditex had increased shipments to Spain by 37% relative to the year before and the fly route was between India and Spain. It was not a one-off rescue mission to empty a full port but can instead be described as the scheduled breach of normal cost limits to continue the flow of new products as competitors lingered on slow boats.

Metric	Standard Strategy (Sea Freight)	Crisis Response (Air Freight)	Strategic Implication
Transport Mode	Maritime via Suez Canal	Air Cargo (Charter & Commercial)	Prioritizing speed and availability over logistics cost efficiency.
Transit Time	25–30 Days (Pre-Crisis) / 40+ Days (Crisis)	< 48 Hours	Decoupling lead times from maritime bottlenecks to maintain freshness.
Volume Change	Decreased flow due to bottlenecks	+37% increase in air consignments	Aggressive capacity booking to secure inventory flow against market shortages.
Cost Implication	Low (\$)	Very High (\$\$\$\$)	Costs absorbed by high gross margins (57.8%), leveraging financial health as a weapon.

Emission Impact	Low Carbon	~35x higher CO2per kg	Direct strategic conflict with long-term Scope 3 sustainability targets.
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Table 4.1: Logistics Modal Shift Analysis (Inditex India-to-Spain Route, 2024)

(Source: Synthesized from Reuters Trade Data 4, Vitality Carbon Studies 6, and Inditex Financial Reports 7)

In its decision to fly its merchandise, Inditex significantly increased the distance between itself and H&M practically within a day. As H&M boxes were slowly sailing around the Cape of Good Hope, leaving European racks bare after taking out spring jackets and flowery dresses in their bags, the Zara packets were landing right on time in Spain. Shoppers who found nothing but bare rails at competing chains simply walked next door to Zara and walked out with shopping bags. What looked like an extravagant freight bill suddenly paid for itself in free publicity and extra sales. The entire trick worked in that Inditex maintains an extremely lean global logistics infrastructure: when pallets are received at Zaragoza, they are resorted to, repackaged, and shipped back out again, direct to stores in Tokyo to Toronto, and back to Dubai to keep the rest of the world equally supplied.

4.3 Procurement and inventory: The shift to "Just-in-Case"

Accounting records of FY2024 indicate that by January 31, 2025, the Inditex inventory increased by 12 percent of the previous year. Furthermore, this increase was the result of a calculated, defensive decision to hold Safety Stock rather than a sign of poor sales performance—revenue increased by 10.5% in constant currency during the same period.

Two important areas were where this shift to a "Just-in-Case" procurement approach was seen:

1. Early deliveries of raw materials: Inditex accelerated the procurement of raw material suppliers, especially the Asian greige (undyed cloth) suppliers. Even if shipment delays were worse, the corporation made sure that its production base would not operate without inputs by importing these products into European proximity hubs.
2. Basic Goods Buffering: Inditex provided greater stock in case of "Functional" products of longer life span, such staples, denim, and conventional menswear. This ensured that stores in the case of occasional inconvenience to trends products flow, were also appealing and profitable.

4.4 Customer service and the unexpected resilience: Integrated Stock Management System (SINT)

SINT enabled Inditex to circumvent these bottlenecks in the presence of the 2024 crisis. If there is a specific delay in transit to the central warehouse but on a rack in a

store in Paris, the algorithm of the system would send an online order made by a buyer in France to the store. The item would be picked by the store staff on the floor, packed, and shipped to the customer. This ability was successful in opening millions of units of stock that would have been idle and languished in the final mile of the chain.

SCM Dimension	Pre-Crisis State	Crisis Response State (2024/25)	Performance Outcome
Procurement	Lean, JIT, Minimal buffers	Strategic Buffering: +12% Inventory levels	Prevention of production stoppages in proximity factories despite Asian delays.
Sourcing Mix	Balanced Asia / Proximity	Heavy Reliance on Proximity: Turkey/Morocco ramped up	Maintained 2-week design-to-shelf cycle for trend items.
Logistics	Cost-optimized Sea/Road	Agility-optimized Air/Road: High cost acceptance	Gross Margin maintained at 57.8% despite massive logistics spend.
Customer Service	High availability	SINT Optimization: Store stock used to fill online gaps	Online sales grew +12% despite logistics chaos; minimal customer friction.

Table 4.2: Impact on SCM Dimensions (2024-2025 Analysis)

(Source: Synthesized from Inditex FY2024 Results 7 and Market Analysis)

5. Recommendations

To be competitive in an environment of volatility and permacrisis, Inditex must do more than abandon its current supply chain system. The recommendations below aim to address the conflict between environmental responsibility, operational agility, and the development of long-term resilience. These methods indicate some paradigm shifts in sourcing and logistics networks to go beyond a mere increase in efficiency.

5.1 Strategic disengagement through a green sea–air transport domain.

The holing in the Red Sea revealed the weakness of a two-polar logistical strategy that leaves rapid high-carbon air freight or slow low-carbon shipping via the port as the only alternatives. Inditex requires a middle ground that will not have so significant negative effects on the environment, and yet should be as fast as air transportation.

Recommendation: Inditex uses Dubai (Jebel Ali Port and Al Maktoum International Airport) as a central multimodal hub, therefore, it should consider creating a SeaAir Hybrid Transport Corridor as the basis of its logistics network.

Operational Mechanism: With this model, clothes produced in South Asia (India, Bangladesh, Pakistan) would not be shipped directly to Spain, which is a very long route that would reduce the highest number of emissions. Instead, they would be shipped by sea to the Jebel Ali Port of Dubai.

For the last, shorter trip to Inditex's logistical facilities in Zaragoza or Barcelona, the products will be transferred to air cargo.

Partnership Integration:

Additionally, this approach needs to be combined with Inditex's current ECO Delivery Ocean program with Maersk. For the maritime part of the Sea-Air corridor, Inditex might further reduce carbon intensity by using green methanol-fueled vessels, establishing a "Green Corridor" that strikes a compromise between sustainability and speed. This may make Inditex qualified for future green logistics incentives or lower CBAM obligations since it is complying with the "Green Corridor" ideas supported by European logistics frameworks.

5.2 Nearshoring 2.0: Managing concentration risk and geographic diversification

The investigation identifies a possible concentration risk, even if proximity sourcing in Turkey and Morocco was crucial during the Red Sea crisis. Inditex's "Responsive" network is mostly dependent on Turkey, which is experiencing economic turbulence of its own, including hyperinflation and currency instability. An excessive dependence on a single nearshore hub creates a single point of failure.

The case of Turkey vs Egypt: The recent reports on the industry reveal that Inditex has reiterated that it is committed to Turkish suppliers and also has no plans to further extend sourcing in Egypt at present. The ties that were established and the infrastructure that have been developed in the Turkish textile industry appear to be the forces behind this decision. Nonetheless, there is a strategic threat associated with this stance. This is because the situation lacks diversity putting the supply chain at risk to regional geopolitical shocks or economic crises in Turkey, but Turkey is significant to the supply chain.

Recommendation: Inditex should consider a Nearshoring 2.0 model that extends Eastern Europe (Romania, Bulgaria) in search of opportunities. The Rationale for Rail Connectivity in Europe: Romania and Bulgaria are connected to Western Europe by the continental rail network, in contrast to Turkey or North Africa, which still need a

road-ferry or sea crossing to get to the Spanish distribution centers (and are also almost vulnerable to maritime delays).

Real Options Application: Establishing supplier clusters in these areas generates "options" to switch production when viewed through a view of Real Options Theory. Inditex may choose to boost volume in Romania if the Bosphorus strait becomes a bottleneck or Turkish inflation makes production unfeasible. This is about building a portfolio of nearshore options that may be adjusted in response to real-time geopolitical and economic circumstances, not about leaving Turkey.

5.3 Digital twin technology for proactive risk management

Although it was reactive, Inditex's approach to the 2024 issue was effective. The over dependence on air freight and inventory buffering was a reaction to an issue that has already arisen. Reactive strategies will result in decreasing benefits and increased expenses as supply chain disruptions increase in frequency.

Recommendation: Inditex makes an investment in digital twin technology in order to build a realistic model of its worldwide supply chain.

The Solution: A digital twin maps every factory, warehouse, port, and transport route in real time, creating a virtual duplicate of the actual supply chain. Contrasting to conventional tracking systems, a Digital Twin simulates future events using artificial intelligence.

Scenario Planning: Inditex may simulate a scenario of a Suez Closure six months ahead to experiment with the various reactions. The system would measure the effect of switching to air freight, sourcing through other hubs, and increasing the safety stock.

Impact Quantification: The Digital Twin would offer SKU-level granularity and understand which products ("red floral blazers") would need to be flown to satisfy the demand, and which could withstand a 10 day delay. This eliminates the blanket deployment of air freight and can be adapted to a surgical logistics response, which costs less money and carbon.

Predictive Maintenance: Not just in logistics, Digital Twins can be used in manufacturing factories near each other (through IoT) to predict a malfunction that will lead to a stoppage of production.

The fifth point, which works to reduce reverse logistics, is localization of circularity. Circular economy, and reverse logistics of returns is also one of the significant hidden weaknesses of Inditex in resiliency. Today, when products are returned, they are often shipped long distances to get processed, sorted, and resold because of the centralization of the logistics which generates numerous Scope 3 emissions.

Recommendation: To ensure faster localization of their operations in the circle, Inditex needs to expand the Zara Pre-Owned platform and transform flagship stores into Circular Processing Centers.

Operational Shift:

US growth: Inditex has introduced the Zara Pre-Owned platform in the United States since it was successful in Europe. This program must be vigorously extended to all the large markets.

Local Resale Mechanism: Inditex should also consider using Pre-Owned to repurpose its returned items locally instead of shipping them back to central warehouses in Spain. The jacket that was sent back to New York must be checked, washed, repackaged, and sold to a spot in New York through the peer-to-peer or shop-controlled system.

In-store repair: This process will decrease the replenishment inventory and will in effect reduce the strain on the upstream supply chain.

6. Conclusion

The objective of this report was to carry out a strategic analysis of Inditex's supply chain resilience to understand whether its successful hybrid sourcing model is a sustainable competitive advantage in the era of geopolitical permacrisis and pressing climate requirements. The Way Ahead: Inditex is also at a significant stage of strategy. To keep its leadership up to 2030, it has to shift its strategy that is currently based on rapid, high-emission growth to sustainable resilience. Institutionalizing Sea-Air corridors through Dubai, nearshore production development to rail-linked Eastern Europe, Digital Twin simulation, and location of circularity are the recommendations in the report compared to simpler operational modifications. They demonstrate the way the business model has changed considerably. With these improvements, Inditex can detach its emissions from its responsiveness, to become more vulnerable to new geopolitical shocks, and achieve its long-term sustainability objectives. This company demonstrated that they are capable of moving faster than anyone; the next mission is to move to a greater level.

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Declaration of Software Tools

I declare that I used Grammarly, Google Gemini solely for the purpose of proofreading and checking the grammar of my own original writing. The tool was used to identify spelling errors. No text or content within the final submission was generated by AI, all writing is my own work.